

Curriculum Vitae — Minwoo Lee

Minwoo Lee

Bachelor Student, Kangnam University — Yongin, South Korea

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github.com/minuum

Portfolio: [minuum-portfolio](https://minuum-portfolio.github.io) | Research Page:

minuum.github.io/EdgeGroundVLA

EDUCATION

Kangnam University — Yongin, South Korea

Bachelor of Science in Artificial Intelligence (Senior, 1st Semester)

- Expected Graduation: Feb. 2027

- GPA: **4.27 / 4.5** (Ranked top of the department)

RESEARCH INTERESTS

- Vision-Language-Action (VLA) Models for Robotics
- Embodied AI & Mobile Robotics & Autonomous Driving
- Sim-to-Real Transfer & Physical Latency Mitigation
- Deep Reinforcement Learning

PUBLICATIONS & MANUSCRIPTS

- **EdgeGround-VLA: On-Device Goal-Directed Navigation via Dual Frozen Grounding Encoders**
Journal of The Korea Society of Computer and Information (JKSCI), KCI Registered, **Under Review (2026)**.
Minwoo Lee (1st Author / Project Lead), et al.
 - Fuses dual frozen grounding encoders (OWLv2 + Kosmos-2) with a 0.866M-parameter action head (1.128M total trained parameters).
 - Achieves **95.0% real-robot goal-reaching success rate** fully on-device on Jetson Orin NX.
 - Project Page: minuum.github.io/EdgeGroundVLA
- **Designing Conversational AI Services Using RAG-Based Loneliness Analysis for Koreans**
Journal of the Korea Convergence Society (JCCT), KCI Registered,

2024.

Minwoo Lee (2nd Author), et al.

CONFERENCES

- Presented at the International Conference on IPACT 2024 —
Outstanding Paper Award
Minwoo Lee (2nd Author), et al.
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RESEARCH EXPERIENCES

Undergraduate Research Assistant — Advised by Prof. Inyeop Choe,
Kangnam University

Project: On-Device Vision-Language-Action Models for Mobile Robotics |
Sep. 2025 – Present

- **EdgeGround-VLA Architecture (1st Author):** Proposed and implemented EdgeGround-VLA, an on-device navigation model fusing heterogeneous vision grounding encoders (OWLv2 + Kosmos-2) with a lightweight 3-DoF action head (0.866M parameters).
 - **Parameter Efficiency & On-Device Deployment:** Engineered a lightweight policy training only 1.128M parameters out of 459M total parameters, achieving 95.0% goal-reaching success rate in 100 real-robot trials on NVIDIA Jetson Orin NX without cloud connectivity or pre-built infrastructure. (*Submitted & Under Review at JKSCI, KCI-registered journal*)
 - **VLA Backbone Analysis & Pipeline Optimization:** Conducted per-layer attention measurement and frozen linear probe classification to analyze VLA backbone representations, developing a 2-stage decomposition pipeline that improved closed-loop navigation performance.
 - **Real-Robot Control Stack:** Configured Jetson Orin Linux and ROS2 driver nodes on the 3-axis omnidirectional Serbot 2 mobile platform; built a 10Hz asynchronous action re-issuance and Jitter Hold filtering stack to maintain continuous control under edge inference latency (~1.02s).
 - **Award:** Won the **Silver Prize at the 2026-1 Kangnam University Capstone Design Competition** during the VLA research project.
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AWARDS AND HONORS

- **Silver Prize**, 2026-1 Kangnam University Capstone Design Competition
- **Grand Prize (Dean of College of Engineering Award)**, 2025 Kangnam University Hackathon 'Kangnengthon' (Jointly hosted by Kangnam University & GDGoC: Kangnam University)
- **Outstanding Paper Award**, IPACT 2024 International Conference
- **First Prize (Best Project Award)**, Software Academic Festival (Kangnam University, 2024)

- **Excellence Award**, AI Academic Festival (Kangnam University, 2024)
 - **Award**, K-HTML Hackathon (2024)
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PROJECTS

Serbot 2 Robot Platform Integration — Individual Project | *Jan. 2025 – Mar. 2025*

- Configured a Jetson Orin-based Linux environment and integrated ROS2 driver nodes for the Serbot 2 omnidirectional mobile robot platform. - Implemented core locomotion control scripts and configured the hardware-software communication interface to host VLA inference models. - Resolved physical-to-virtual command delays and stabilized actuator control signals.

Ladi — Software Academic Festival | *Oct. 2024*

- Developed an AI-based healthcare routine recommendation system (**Ladi**), winning the **First Prize**. - Implemented a hybrid search system combining dense (FAISS) and sparse (KiwiBM25) retrievers to optimize medical QA search. - Integrated a CrossEncoder Reranker to improve recommendation quality, and optimized prompt templates using LangSmith. - Preprocessed AIHub healthcare QA data to categorize diseases and route intents.

Liberty — AI Academic Festival | *May 2024*

- Served as Project Manager (PM) and lead architect for an AI Legal QA Agent system (**Liberty**), winning the **Excellence Award**. - Built a multi-agent workflow using LangGraph, integrating dynamic query rewriting and quick filtering. - Implemented a hybrid retriever (FAISS + KiwiBM25 + Pinecone) achieving 1.000 score in faithfulness and retrieval accuracy benchmarks.

SKILLS AND TECHNIQUES

- **AI/ML**: PyTorch, HuggingFace, LoRA/PEFT, PaliGemma2, CLIP, SigLIP, Kosmos-2, LangGraph (Agent), LangSmith, FAISS, Pinecone (Vector DB), BM25, CrossEncoder (Reranker)
 - **Robotics & Infra**: ROS2, Jetson Orin, Linux (Ubuntu), Locomotion Control, Asynchronous Control, Git
 - **Programming Languages**: Python, C++, SQL
 - **Languages**: Korean (Native), English (Intermediate — preparing for TOEIC)
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COURSEWORK

- **Deep Reinforcement Learning** — *Self-study (UC Berkeley CS285: Deep RL course lectures & homework)*
 - Focused on Imitation Learning, Policy Gradients, Actor-Critic methods, and value-based RL.